MIL-W-12307C(EA)
16 June 1987
SUPERSEDING
MIL-W-12307B(EA)
21 November 1977

MILITARY SPECIFICATION

WRENCHES, VALVE-REMOVING, M1 AND M2

This specification is approved for use within the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of wrenches.

1.2 Classification. The wrenches shall be of the following types (see 6.1 and 6.2):

Type I - M1 Wrench
Type II - M2 Wrench

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5120

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PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.

MILITARY

MIL-P-116 - Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

68-6-2 - Wrench, Valve-Removing, M1.
68-6-3 - Wrench, Valve-Removing, M2.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D 3951 - Commercial Packaging (DOD adopted).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 68-6-2 or 68-6-3 shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the wrenches shall conform to the specifications and drawings listed on Drawing 68-6-2 and 68-6-3.

3.2 Assembly.

3.2.1 M1 wrench. The wrench shall be assembled in accordance with Drawing 68-6-2.

3.2.2 M2 wrench. The wrench shall be assembled in accordance with Drawing 68-6-3.

3.3 Workmanship. The wrenches shall be free from foreign matter (oil, grease or dirt) and damage such as cracks or deformation which could affect their intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the wrenches produced by one manufacturer, at one plant, from the same materials and under essentially the same manufacturing conditions.

4.2.2 Sampling for examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination. The sample wrenches shall be examined in accordance with the classification of defects (4.2.3.2) and MIL-STD-105.

4.2.3.2 Classification of defects.

(a) Wrench, valve-removing, M1 and M2 (Drawing 68-6-2 and 68-6-3).

Categories		Defects
Critical:		None defined
Major:	AQL	1.0 percent defective
101		Set screw missing
102		Wrench opening, size incorrect
103		Guide size and location incorrect
104		Workmanship
Minor:	AQL	2.5 percent defective
201		Handle length incorrect
202		Body diameter incorrect
203		Wrench flat, thickness and overall width incorrect
204		Protective finish missing
205		Chamfers missing
206		Hub diameter incorrect

5. PACKAGING

5.1 Cleaning. The wrenches shall be cleaned in accordance with process C-1 of MIL-P-116 and dried by any procedure listed therein, as applicable.

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5.2 Unit packing.

5.2.1 Level A. The wrenches, cleaned in accordance with 5.1, shall be unit packed, one per box, in fiberboard boxes conforming to style optional, grade W5c and V3c, class weather-resistant of PPP-B-636. The handle shall be disassembled from the hub, the threaded areas coated with type P-9 preservative in accordance with MIL-P-116 and, with the set screw screwed into the hub section reasonably tight, the two sections shall be placed into the box, separated and restricted from free movement by cushioning or blocking with appropriate weather-resistant material such as fiberboard. The boxes shall be closed and waterproofed by sealing the seams and joints with tape in accordance with the appendix to PPP-B-636.

5.2.2 Level B. Unit packing shall be in accordance with 5.2.1 except the box and cushioning material may be domestic class and waterproofing is not required.

5.2.3 Commercial unit packing. Commercial unit packing shall be in accordance with ASTM D 3951.

5.3 Packing.

5.3.1 Level A. The wrenches, unit packed in accordance with 5.2.1, shall be packed 10 per box in a wood box conforming to either style 4, class 2 of PPP-B-621, grade A requirements, or PPP-B-601, overseas type, grade A requirements. The boxes shall fit the contents snugly. Fiberboard filler pads of weather-resistant material conforming to PPP-F-320 shall be utilized as necessary to restrict free movement of the unit packs. The boxes shall be closed and strapped in accordance with the appendix to the respective box specification. Strapping shall conform to finish B of QQ-S-781.

5.3.2 Level B. The wrenches, unit packed as specified, shall be packed 10 per box, in wood boxes conforming to either style 4, class 1 of PPP-B-621 or domestic type PPP-B-601. The boxes shall fit the contents snugly. Domestic class fiberboard filler pads shall be utilized as necessary, to restrict free movement of the unit packs. The boxes shall be closed and strapped in accordance with the appendix to the respective specification.

5.3.3 Commercial packing. Commercial packing shall be in accordance with ASTM D 3951.

5.4 Marking.

5.4.1 Marking, level A and B. In addition to any special marking specified in the contract or order, marking shall be in accordance with MIL-STD-129.

5.4.2 Marking, commercial. Commercial marking shall be in accordance with ASTM D 3951.

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6. NOTES

6.1 Intended use. The M1 wrench is intended for use in removing or replacing 1-inch angle valves from 1-ton containers. The M2 wrench is intended for use in removing or replacing Chlorine Institute type valves from 1-ton containers.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether type I or type II wrenches are required.
(c) Title, number and date of applicable drawing.
(d) Level of unit packing and packing required (see section 5).

6.3 Subject term (key word) listing.

Angle valve
Chlorine Institute type valve
Valve
Wrench

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

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Preparing activity:

Army - EA

Project No. 5120-AC16

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-12307C(EA)	WRENCHES, VALVE-REMOVING, M1 AND M2

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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